A Case Study to Make Use of Cross-channel User Experience for Interactive Exhibition Design

Simge Esin Orhun
Dept.of Communication Design
Faculty of Architecture and Design
Ozyegin University, Istanbul - Turkey

1. Introduction : The problem

Processing of information

interacting + experiencing : learning

storytelling and information technologies

MUSEUMS:
environments to experience

Evolution

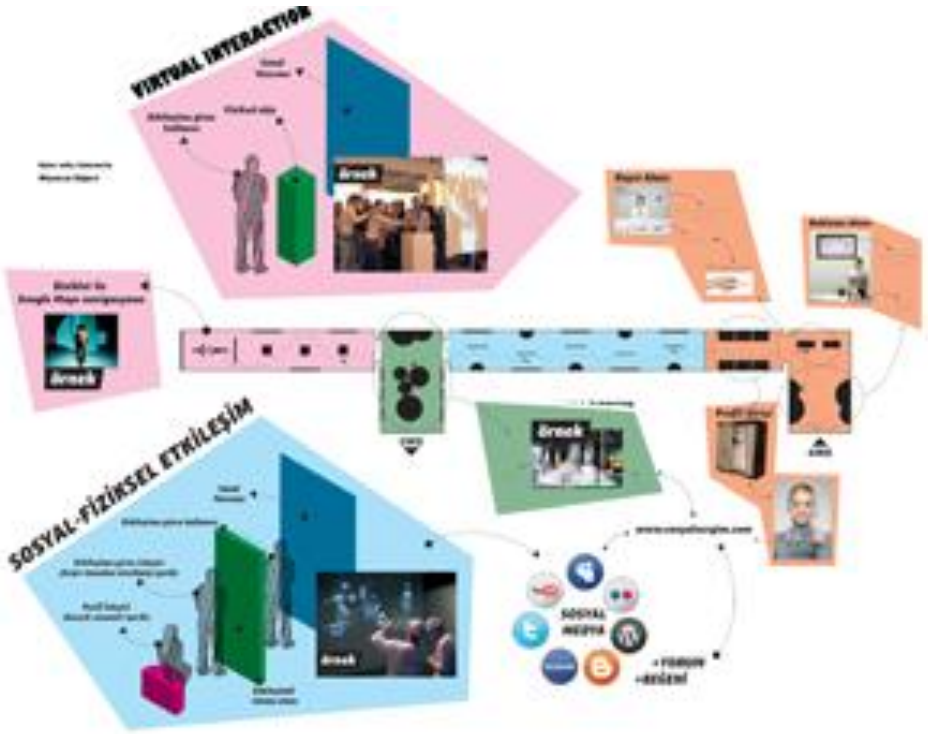
Multitouch displays terminals	dissemination of information	INTERACTIVE
interactive installations	kinaesthetic learner (learn by doing)	PARTICIPATORY
		ENGAGEMENT?



1. Introduction : The problem

Interactive exhibiting is an area that aims the visitor to go beyond being the participant and acquire the role of the actor and <u>contribute directly for the realization of the exhibition himself.</u>

Technical qualities? Variety of interaction modes, displaying techniques were studied.
No individualistic contribution to the theme. To dependent on technology, like a theme-park.





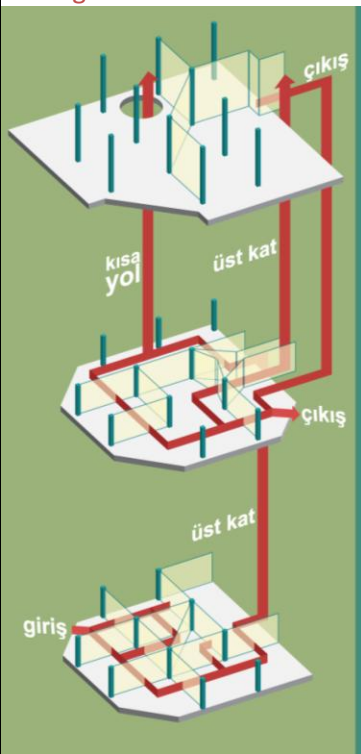
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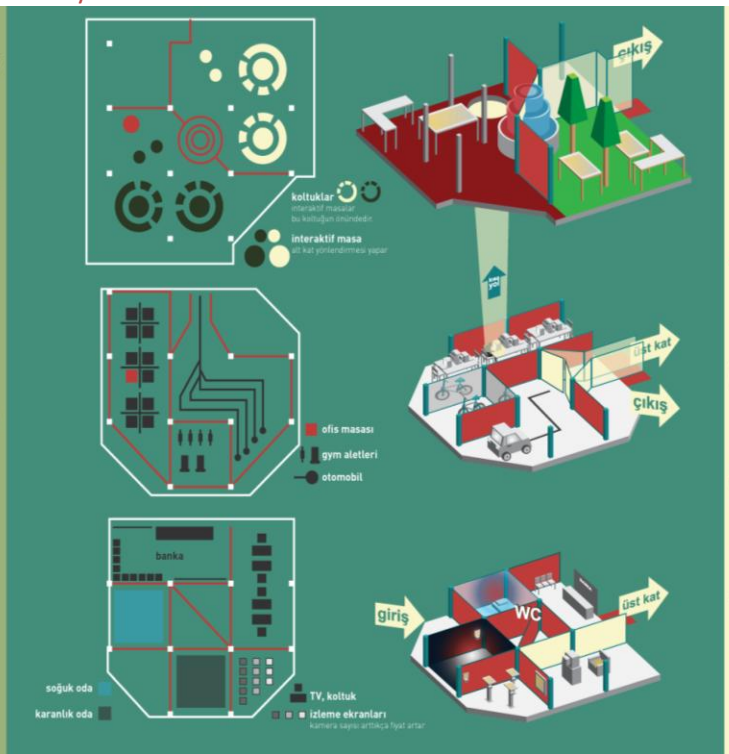
Technical qualities

Functional qualities? Storytelling and scenography has been studied in parallel with the understanding of architectural space. Multistory spaces mostly failed to keep the unity of the theme and the sustainability of the exhibition.

Navigation



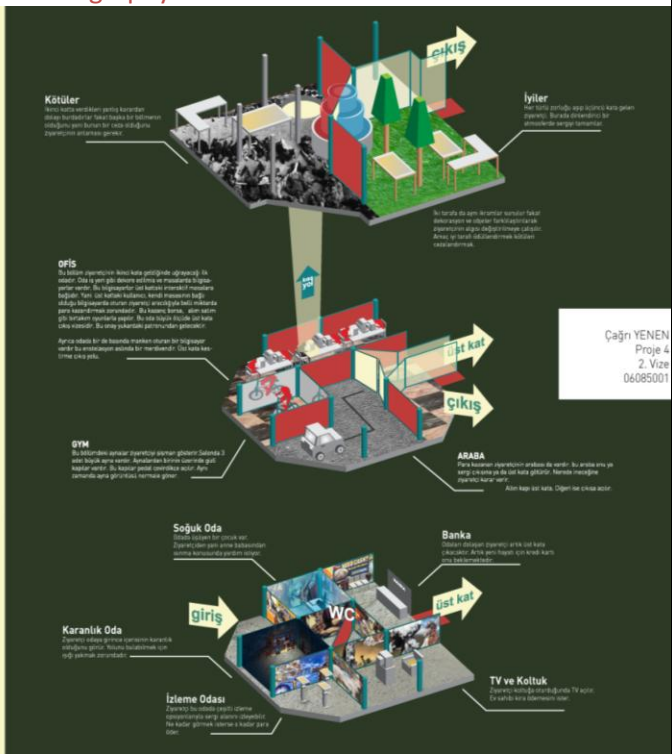
layout



interaction



scenography



1. Introduction : The aim

AIM: to provide an alternative structural design for the shared information that would enhance the engagement of the visitor in terms of the behaviorial aspects

storytelling and information technologies

MUSEUMS:
environments to experience

Evolution

displays Multitouch terminals	dissemination of information	INTERACTIVITY	technical
interactive installations	kinaesthetic learner (learn by doing)	PARTICIPATORY	functional
?	?	ENGAGING	behavioural aspects

2. User experience to combine technical, functional and behaviorial aspects

Technical qualities? Too much focus on technology, loss of attention and connection to the theme. Fragmentation.

Functional qualities? Spatial unity is lost in multistory spaces and in linear spaces divided into rooms. Fragmentation.

Behaviorial qualities?

How can we decrease fragmentation?

1. STRATEGY: USER EXPERIENCE.: visitor experience. Approach form the point of COGNITIVE Psychology.

In order for the exhibition space to be an engaging place, the behaviorial patterns (memories, experiences, ... etc.) associated with the local has to be integrated within the design process.

2. STRATEGY: Give up on multi-channel traditional strategies and started studying cross-channel user experience.

Definition of cross-channel user experience: « a single service to be spread across multiple channles in such a way that it can be experienced as a whole only by interacting with a number of media.»

.

3. Method For the Research

Interactive exhibition > an engaging space

Technical aspects > interactive technologies
Functional Aspects > architectural space (local)
Behaviorial aspects > user experience

Cross-channel user experience

3. STRATEGY: **RULE BREAKING METHODS >**

THE OBSTRUCTION METHOD: placing obstacles in the users way.

Idea of creating an exhibition which assumed the users' one of the senses to be disabled as a result of the environmental factor.

Spatial Interaction Design Studio
2012-2013 term
12 undergraduate students - prior to graduation project

<u>Steps of the design process:</u>

1. Background research for interactive exhibiting
2. A review of technological platforms
3. General tendency of the design : visitors to be disabled by the environment and visitors to learn by experiencing
4. Creating a theme and a storytelling
5. developing preliminary sketches and layouts
6. Structuring the interaction scenarios in terms of activities, actions and operations.

4. evaluation

✓ successfull solutions to connect the spatially distant and separated interactions and experiences

Being disabled from one of his senses, the visitor navigates in the exhibition through experiencing.
While the visitor is doing and interacting, he is learning about the theme.
He becomes the actor who will have his own interpretation of the theme.

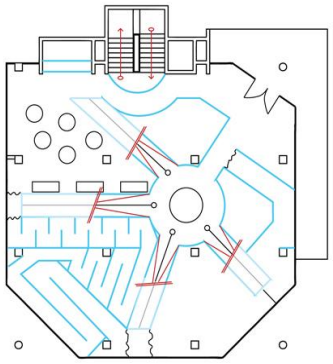
In order to prevent the fragmentation in the unity of storytelling and spatial experience; a mobile or a wearable tool is added to the exhibition, so as to connect each spatially distant and seperate interaction.

These tools were planned
. to **carry information** concerning the progress of the visitor in the exhibition,
. to **guide the navigation** within the space
. to **privatize the experiences** of each visitor.
. To provide **connection to social media** when needed,
. To archive **each visitor's experience**

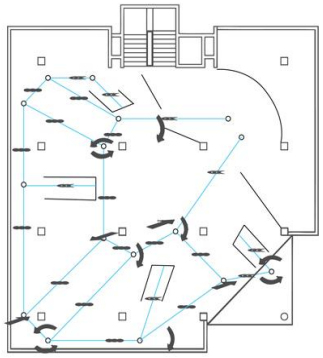
i. Disabled for seeing «sense-i : master the experience and sharpen your senses»
focused on the learning experiences in a blind environment and
aimed to enhance the visitor's way finding capability by improving his sense of touching

4. Evaluation

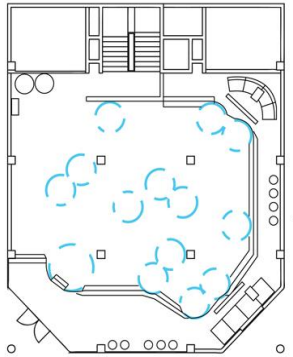
sense·i master the experience and sharpen your senses



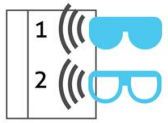
Ground Floor plan -
learning by experience



First Floor plan -
the layout of the activities



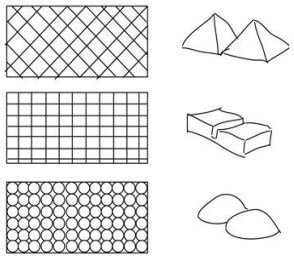
Basement Floor plan



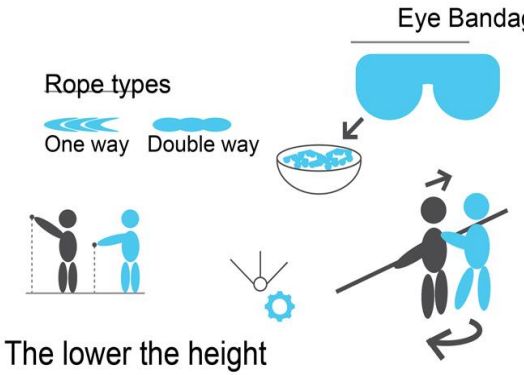
eye bandage used together with eye glasses



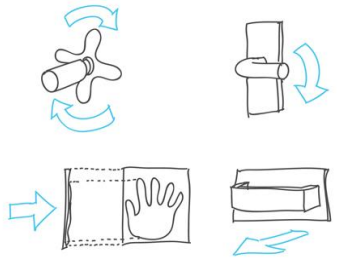
Personal Placement mark



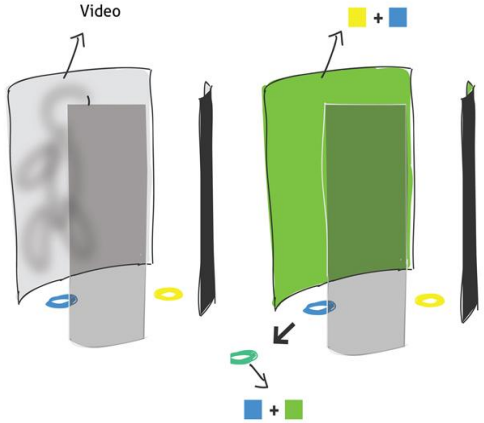
Learning experience by touching different materials



The lower the height of rope, the closer you are to the end.



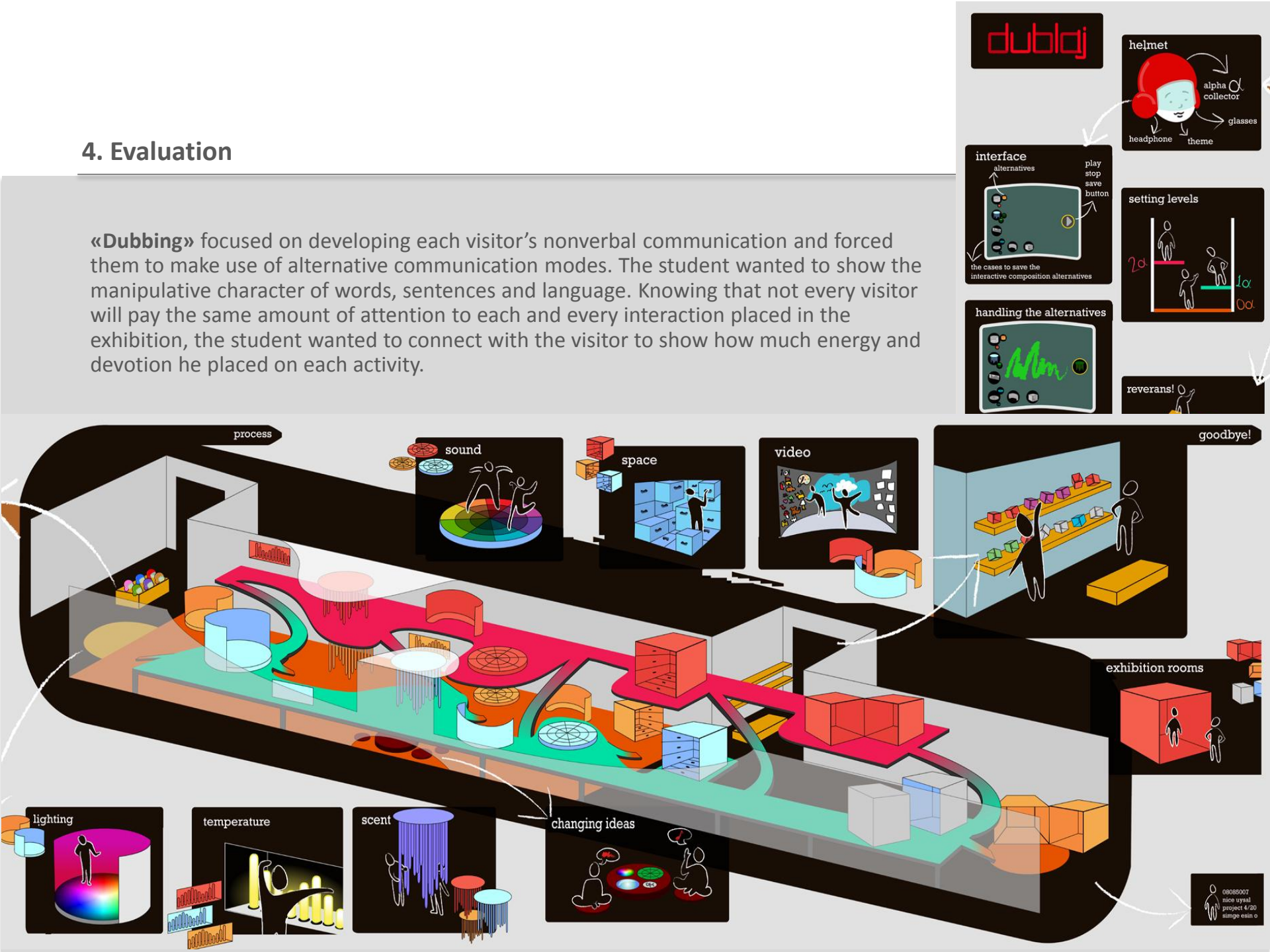
On the junctions, use one of four of the analogue interaction mechanizms to select your route.



Color of the panels change, indicating the users rroute

4. Evaluation

«Dubbing» focused on developing each visitor's nonverbal communication and forced them to make use of alternative communication modes. The student wanted to show the manipulative character of words, sentences and language. Knowing that not every visitor will pay the same amount of attention to each and every interaction placed in the exhibition, the student wanted to connect with the visitor to show how much energy and devotion he placed on each activity.



«House of Media» providing the user to experience new media consumption. Takes place in the home as the fundamental interaction environment. Visitor has a seed to grow a plant. From the moment, the user begins to make data entry, the plant starts to grow. Excessive and inconsisten data make plant weak, proper data make plant live healty.

4. Evaluation

[HOUSE OF MEDIA]

Medya Evi is an interactive exhibition space design project related to relationship between human and the new media.

Medya Evi / House of Media

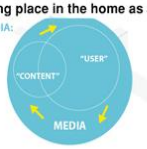
Concept: "Content"

Keywords: Content:up-to-date data, agenda, news, statements...
User: The person who uses , shapes and consumes the content
Media: The system putting both user and content together

Chosen Venue: Yuksel Sabanci Arts Center
Target Group: Internet users 25 years and older.

Aim: Providing the user to experience new media content consumption which is taking place in the home as a fundamental interactive enviroment.

MEDIA:



METAPHOR: PLANT

From the moment user begins to make data entry, the plant starts to grow.

Excessive and inconsistent user data makes the plant weak. If user doesn't make any data entry for a long time, the plant dies.

If the contents include proper and consistent data, the plant continues to live healthy and grows.

Additional device: Flashlight



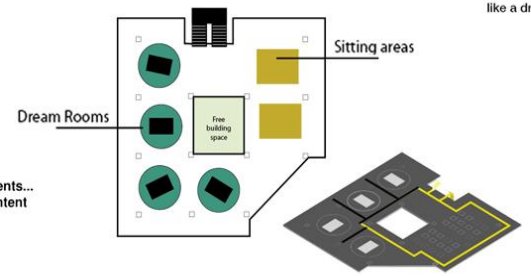
Stores the data in its memory

Displays the datas visually like a projection

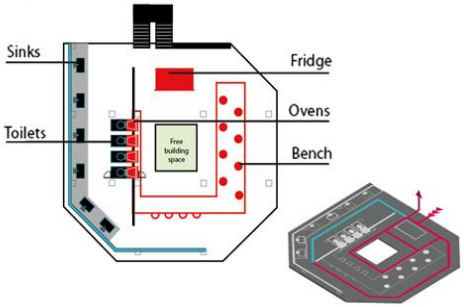


FLOOR PLANS

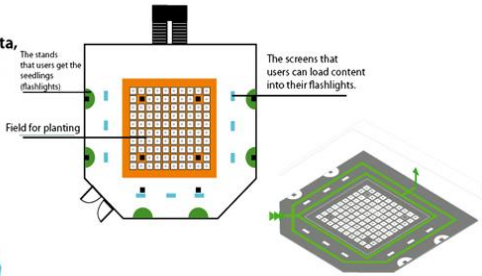
3. Floor : Living and Resting



2. Floor : Basic Needs



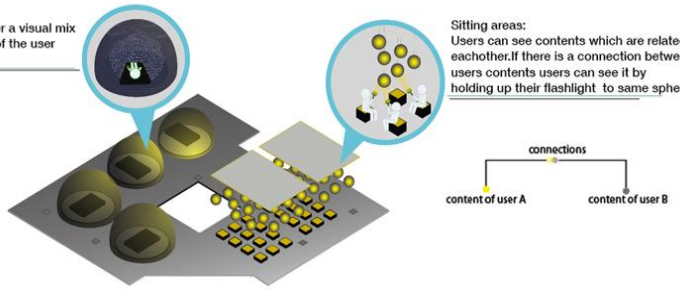
1. Floor : Garden



SPATIAL INTERACTIONS

Dream Rooms:
These rooms show to user a visual mix of the whole experience of the user like a dream.

Sitting areas:
Users can see contents which are related to eachother.If there is a connection between users contents users can see it by holding up their flashlight to same sphere.

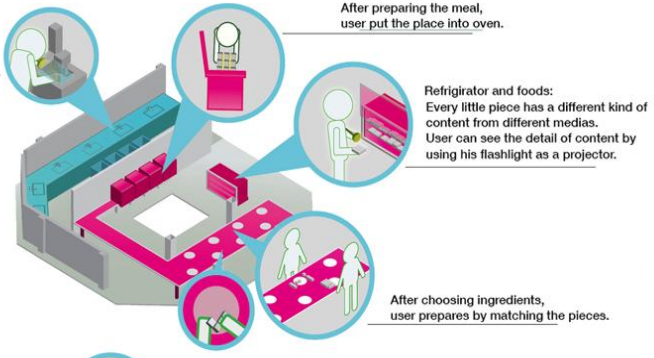


This sink lets user to see text based datas related to his content.

After preparing the meal, user put the place into oven.

Refrigirator and foods:
Every little piece has a different kind of content from different medias.
User can see the detail of content by using his flashlight as a projector.

After choosing ingredients, user prepares by matching the pieces.

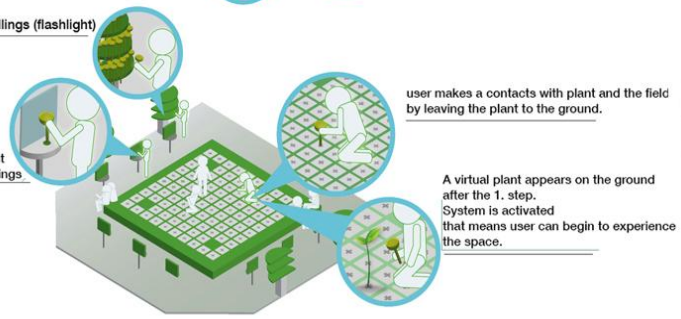


User selects a seedlings (flashlight)

user makes a contacts with plant and the field by leaving the plant to the ground.

User selects a content and save it into seedlings

A virtual plant appears on the ground after the 1. step.
System is activated that means user can begin to experience the space.



5. Conclusion

Among other institutions who are also passing through a transformative period in terms of their functions and roles in society, **museums are also looking for ways to improve their efficiency.**

Throughout the research, solid outcomes were obtained in terms of interactive exhibiting that involved
. the use of architectural means to solve the **functional needs** of the design and
. the interactive tools to match the **technical aspects** of design and,
. the perceptual issues to integrate the **behaviourial aspects** ,
and bring the concept of exhibition design **to a level of maturity**.

Making use of cross-channel user experience was beneficial:
. in providing the designer a holistic approach to maintain both the unity and originality of the exhibition.

. for the visitor, to have make his own personlaized interpretation of exhibition.

. in terms of technological use, to give rise to the invention of new tools which can be integrated suitable for decent purposes.

5. Conclusion

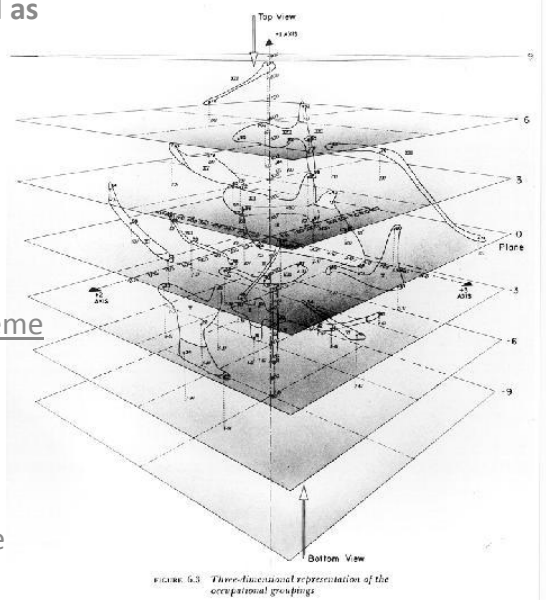
It is possible to say that the exhibition space has huge potential to be perceived and designed as a layered artifact, involving functional, technical and behavioral aspects

«No need to rely on the technological innovations or a specific spatial geometry»

This brings us closer to the idea of a constructivist sense of meaning-making:

It brings the possibility of both externalization of information by the structural design of the theme and the spatial layout of the digital tools together with the internalization of the knowledge obtained by the visitors.

By making use of this approach, it becomes possible for museums to strengten thier roles in the society **as centers for learn-by-experience** and to be one of the **building blocks** of the communication infrastructure of the future cities.



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THANK YOU.

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